

# Scaling Book Exercises: Section 4

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**Chapter:** All the Transformer Math You Need to Know

**Source scan:** ../handwritten/section\_04\_handwritten.pdf

**Notation.**  $B$  is batch size,  $T$  is query length,  $S$  is KV length,  $D = d_{\text{model}}$ ,  $F = d_{\text{ff}}$ ,  $L$  is layer count,  $V$  is vocabulary size,  $N$  is the number of query heads,  $K$  is the number of KV heads,  $H$  is head dimension, and  $G = N/K$  is the number of query heads per KV head. Thus  $NH = D$ .

## Exercise 1

For  $D = 4096$ ,  $F = 4D = 16384$ ,  $V = 32000$ , and  $L = 64$ , the parameter count is

$$P = L(3DF + 4DNH + D) + 2DV.$$

Using  $NH = D$ ,

$$\begin{aligned} P &= 64(3(4096)(16384) + 4(4096)^2 + 4096) + 2(4096)(32000) \\ &= 16,001,335,296 \approx 16.0\text{B parameters.} \end{aligned}$$

Ignoring the small normalization and embedding terms, the attention fraction is

$$\frac{4D^2}{4D^2 + 3D(4D)} = \frac{4}{16} = \frac{1}{4}.$$

With int8 keys and values, the KV cache per token is

$$2LNH = 2LD = 2(64)(4096) = 524,288 \text{ bytes} = 512 \text{ KiB.}$$

## Exercise 2

For  $A[B_X, D_Y] *_D W[D_Y, F]$  on a mesh  $\{X : 4, Y : 8, Z : 4\}$ , the mathematical operation has

$$2BDF$$

FLOPs. Since the computation is replicated across the unsharded  $Z$  axis, the hardware performs

$$2BDFZ$$

FLOPs in total. Each device handles a  $1/(XY)$  shard of the mathematical work, so the per-device work is

$$\frac{2BDF}{XY} = \frac{2BDF}{32} = \frac{BDF}{16}.$$

### Exercise 3

In

$$A[I, J, K, L] * B[I, J, M, N, O] \longrightarrow C[K, L, M, N, O],$$

$I$  and  $J$  are contracting dimensions. Every distinct dimension contributes once, and multiply-add contributes a factor of two. Therefore,

$$\boxed{2IJKLMNO \text{ FLOPs}}.$$

### Exercise 4

Ignoring the Q/K/V/O projections and softmax's lower-order cost, grouped-query self-attention reads Q and the KV cache, performs the  $QK^\top$  and  $AV$  matmuls, and writes the output. In bf16,

$$\begin{aligned} \text{FLOPs} &= 4BTSNH = 4BTSKGH, \\ \text{bytes} &= 4BTN H + 4BSKH = 4BHK(TG + S). \end{aligned}$$

Hence

$$\boxed{\text{AI}(T, S, G) = \frac{TSG}{TG + S}}.$$

During prefill,  $S = T$ , so

$$\text{AI}_{\text{prefill}} = \frac{TG}{G + 1}.$$

On TPU v5e, whose hardware arithmetic intensity is approximately 240 FLOPs/byte, attention becomes compute-bound when

$$T \gtrsim 240 \frac{G + 1}{G}.$$

For MHA,  $G = 1$ , giving  $T \gtrsim 480$ ; for large  $G$ , the threshold approaches 240.

During single-token generation,  $T = 1$ , and

$$\text{AI}_{\text{generation}} = \frac{SG}{G + S} \longrightarrow G \quad (S \rightarrow \infty).$$

Thus generation attention is normally bandwidth-bound because  $G \ll 240$ .

For prefill with  $S = T$ , the attention-to-FFW FLOP ratio is

$$\frac{12BT^2NH}{18BTDF} = \frac{2T}{3F} = \frac{T}{6D} \quad (F = 4D).$$

After accounting for the bandwidth roofline, attention's effective cost is nearly flat while bandwidth-bound, then grows linearly after the crossover.

### Exercise 5

Training FLOPs for the Q/K/V/O projections and self-attention are respectively

$$24BTDNH \quad \text{and} \quad 12BT^2NH.$$

Equating them gives

$$24BTDNH = 12BT^2NH \implies T = 2D.$$

For  $D = 4096$ ,

$$\boxed{T = 8192}.$$

## Exercise 6

If only the seven principal matmul outputs are saved, the backward pass must recompute

$$QK^\top \quad \text{and} \quad \text{softmax}(QK^\top)V$$

to obtain the output-projection gradient. Each matmul costs  $2BT^2NH$ , so the additional matmul work is

$$\boxed{4BT^2NH \text{ FLOPs}}.$$

There are also lower-order elementwise recomputations of order  $O(BTD)$  and  $O(BTF)$ .

## Exercise 7

The H800's FP8 peak without structured sparsity is approximately half of the advertised sparse value:

$$\frac{3026}{2} = 1513 \text{ TFLOPs/s} = 1.513 \times 10^{15} \text{ FLOPs/s}.$$

The available hardware FLOPs over 2.79 million H800-hours are

$$(2.79 \times 10^6)(1.513 \times 10^{15})(3600) \approx 1.52 \times 10^{25}.$$

Using the  $6NT$  training approximation with 37B activated parameters and 14.8T tokens,

$$6(37 \times 10^9)(14.8 \times 10^{12}) \approx 3.29 \times 10^{24} \text{ FLOPs}.$$

Therefore the model FLOPs utilization is

$$\boxed{\frac{3.29 \times 10^{24}}{1.52 \times 10^{25}} \approx 21.7\%}.$$

## Exercise 8

For one int8 expert matrix, loading all experts costs  $EDF$  bytes. If every token selects  $k$  experts, the computation costs  $2kBDF$  FLOPs. Thus

$$\text{AI} = \frac{2kBDF}{EDF} = \frac{2kB}{E}.$$

To exceed TPU v5e's approximate 240 FLOPs/byte roofline,

$$\frac{2kB}{E} > 240 \implies \boxed{B > \frac{120E}{k}}.$$

For DeepSeek with  $E = 256$  routed experts and  $k = 8$ ,

$$\boxed{B > \frac{120(256)}{8} = 3840 \text{ tokens}}.$$